



Basic Principles of Medicine II

Module: Neuroscience & Behavioral Sciences| Lecture NB 22

Histology of the Ear

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Recommended Reading

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 25: Ear

[Ear | Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 9e | Premium Basic Sciences | Health Library \(oclc.org\)](#)

.....
[Article on Histology of the Ear](#)

[Sketch Fab – Bony Labyrinth](#)

Objectives

Code	Objective
SOM.MK.I.BPM2.4.NB.1.HCB.1359	List the overall functions of the divisions of the ear.
SOM.MK.I.BPM2.4.NB.1.HCB.1360	Discuss the divisions of the ear (external, middle and internal) including their structural components.
SOM.MK.I.BPM2.4.NB.1.HCB.1361	State the function of the tympanic membrane.
SOM.MK.I.BPM2.4.NB.1.HCB.1362	Describe the histological structure of the tympanic membrane.
SOM.MK.I.BPM2.4.NB.1.HCB.1363	Compare endolymph vs. perilymph in terms of their composition and location.
SOM.MK.I.BPM2.4.NB.1.HCB.1364	Identify the semicircular canals, vestibule and cochlear canal of the bony labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1365	List the components of the membranous labyrinth (semicircular ducts, utricle and saccule, cochlear duct)
SOM.MK.I.BPM2.4.NB.1.HCB.1366	List the two broad divisions of the membranous labyrinth (vestibular and cochlear)
SOM.MK.I.BPM2.4.NB.1.HCB.1367	List the sensory regions associated with the vestibular part of the membranous labyrinth (crista ampullaris, maculae)
SOM.MK.I.BPM2.4.NB.1.HCB.1368	Describe the histological structure of hair cells associated with the sensory regions of the vestibular division of the membranous labyrinth.
SOM.MK.I.BPM2.4.NB.1.HCB.1369	Describe the two types of hair cells of the vestibular labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1370	Describe the general function of hair cells in the ear including how they convert mechanical vibrations into electrical signals involved in sound perception.

Objectives

Code	Objective
SOM.MK.I.BPM2.4.NB.1.HCB.1371	State the function of the crista ampullaris
SOM.MK.I.BPM2.4.NB.1.HCB.1372	Describe the histological structure of the crista ampullaris
SOM.MK.I.BPM2.4.NB.1.HCB.1373	Explain vertigo resulting from damage to the crista ampullares
SOM.MK.I.BPM2.4.NB.1.HCB.1374	State the function of the saccule and utricle
SOM.MK.I.BPM2.4.NB.1.HCB.1375	Describe the histological structure of the maculae of the saccule and utricle
SOM.MK.I.BPM2.4.NB.1.HCB.1376	Explain vertigo due to the damage to the maculae of the saccule and utricle.
SOM.MK.I.BPM2.4.NB.1.HCB.1377	Give the function of the cochlea duct.
SOM.MK.I.BPM2.4.NB.1.HCB.1378	State the sensory region associated with the cochlea part of the membranous labyrinth (Organ of Corti)
SOM.MK.I.BPM2.4.NB.1.HCB.1379	Describe the two types of hair cells of the cochlear labyrinth.
SOM.MK.I.BPM2.4.NB.1.HCB.1380	Describe the histological structure of the hair cells associated with the sensory regions of the cochlear division of the membranous labyrinth
SOM.MK.I.BPM2.4.NB.1.HCB.1381	Describe the microscopic structure of the cochlea duct.

Objectives

Code	Objective
SOM.MK.I.BPM2.4.NB.1.HCB.1382	State the function of the Organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1383	Identify the location of the Organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1384	Describe the histological structure of the Organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1385	Explain the hearing loss due to damage to organ of Corti
SOM.MK.I.BPM2.4.NB.1.HCB.1386	Explain the modality of sound conduction through the ear

Lecture Overview

Divisions of the ear: Anatomical organization, Histological characteristics, Functional characteristics

- Outer Ear
- Middle Ear
- Inner Ear
 - Bony and Membranous labyrinths
 - Membranous labyrinth
 - Vestibular system - sensory regions and role in perception of movement/balance
 - Cochlear system - sensory region and mechanism of sound perception

3 Divisions of the Ear

1. External Ear

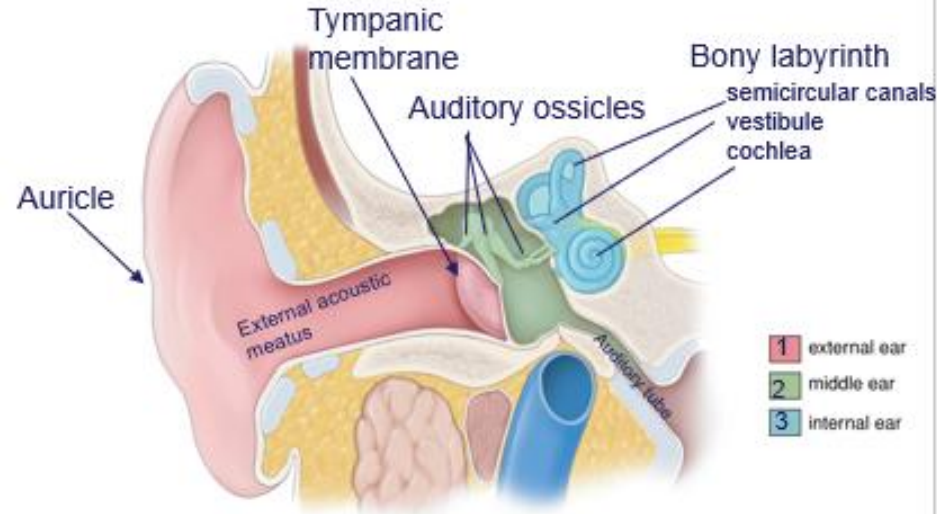
- Auricle (Pinna)
- External acoustic meatus
- Clinical correlation: Conductive hearing loss

2. Middle Ear:

- Tympanic membrane
- Auditory ossicles: 1. Malleus (hammer) 2. Incus (anvil) 3. Stapes (stirrup)
- Auditory tube connects middle ear to the nasopharynx
- Clinical correlation: Conductive hearing loss

3. Internal Ear

- Bony labyrinth
- Membranous labyrinth
- Clinical correlations: Vertigo, Sensorineural hearing loss



Functions:

- **Vestibular system** for balance, posture, and spatial orientation by detecting head movement and changes in position.
- **Auditory system** for sound perception

External Ear

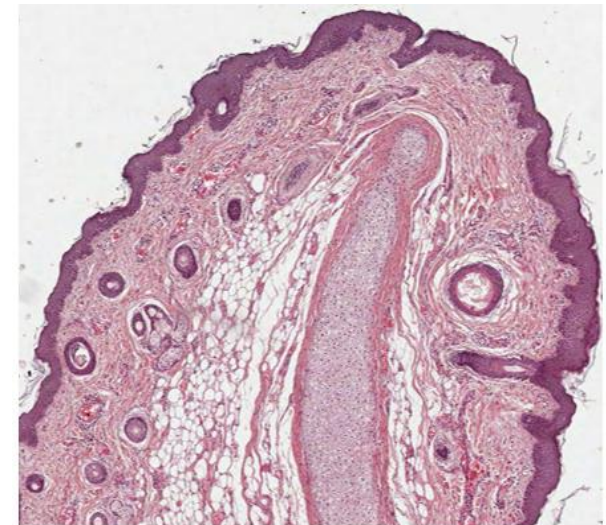
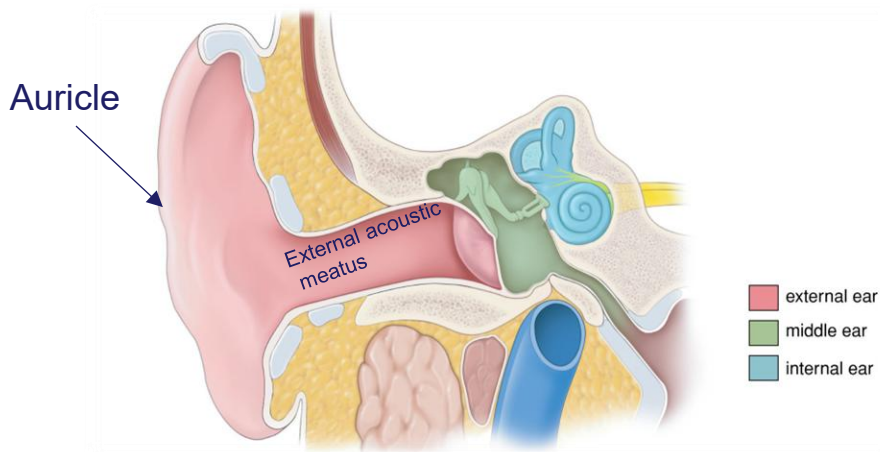
○ Auricle (Pinna)

- Collects & directs sound to the external acoustic meatus

○ External acoustic meatus

- Conducts and amplifies sound to the tympanic membrane
- Lateral 1/3 is cartilaginous (continuous with auricle), lined by skin with hair follicles, sebaceous glands, and ceruminous glands.
- Medial 2/3 is contained within the temporal bone, lined by thinner skin, fewer hairs** and glands.
- **NO ECCRINE SWEAT GLANDS.**

○ Clinical correlation: Conductive hearing loss



Auricle (Pinna)

Epithelium: Stratified squamous keratinized
Type of cartilage: Elastic cartilage

External Acoustic Meatus

What type of glands produce cerumen (ear wax)?

- Ceruminous glands (specialized apocrine glands)

What is cerumen

- Secretion from the ceruminous glands (lipids+ proteins) + desquamated cells + sebum

What are the functions of cerumen?

- Clean/evacuate desquamated cells
- Lubrication
- Impede foreign bodies
- Antimicrobial (slightly acidic)

Middle Ear

AKA Tympanic Cavity

Converts sound waves from the external acoustic meatus into mechanical vibrations that are transmitted to the internal ear.

- **Tympanic membrane**

- Separates the external meatus from the middle ear

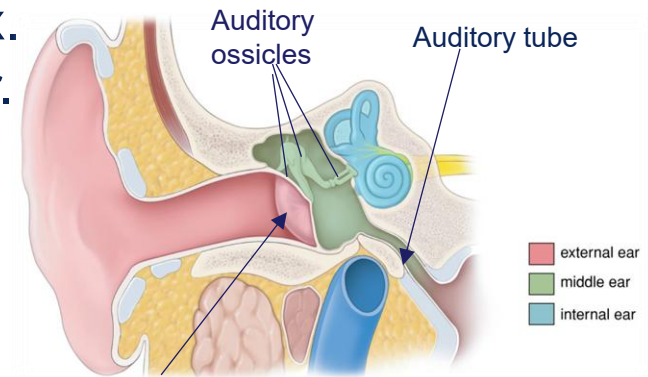
- **Auditory ossicles**

- Connects the tympanic membrane to the inner ear via the oval window.
 - 1. Malleus (hammer) 2. Incus (anvil) 3. Stapes (stirrup)

- **Auditory tube (Eustachian Tube)**

- Connects middle ear to the nasopharynx.
- Equalizes the pressure of the middle ear.

- **Clinical correlation: Conductive hearing loss**

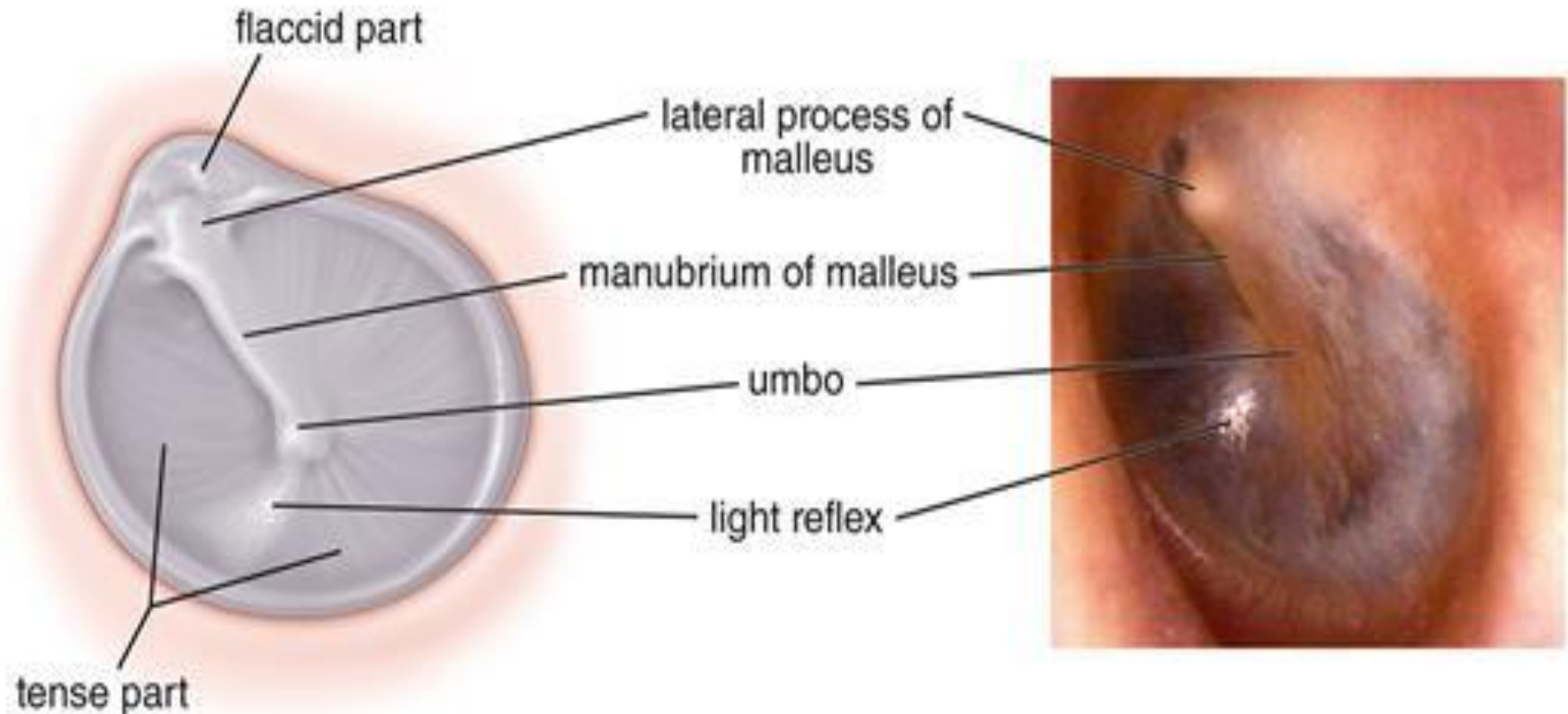


Tympanic membrane

Middle Ear

Tympanic Membrane

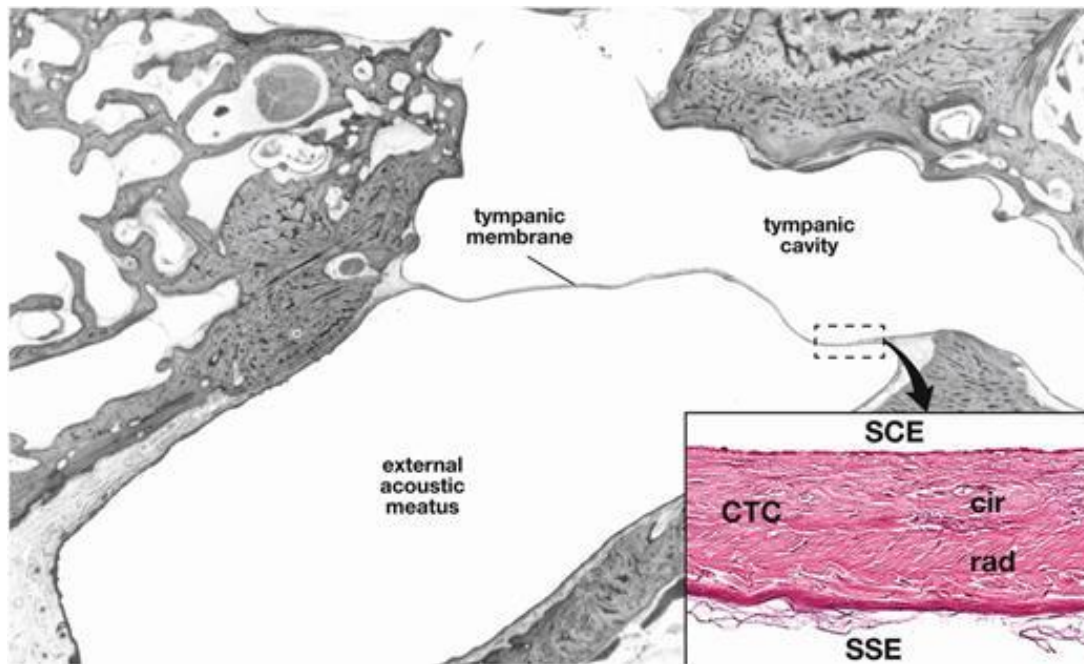
- Semitransparent, thin concave membrane (surface area of 65mm²)
- Forms the medial boundary of the external acoustic meatus and the lateral wall of the middle ear.
 - Starts the conversion of soundwaves to mechanical vibrations.



Middle Ear

Tympanic Membrane

- Three layers:
 - Skin of the external acoustic meatus: **stratified squamous keratinized epithelium**
 - Core of connective tissue: **outer radially and inner circularly arranged collagen fibers.**
 - Mucous membrane of middle ear: **simple cuboidal epithelium.**



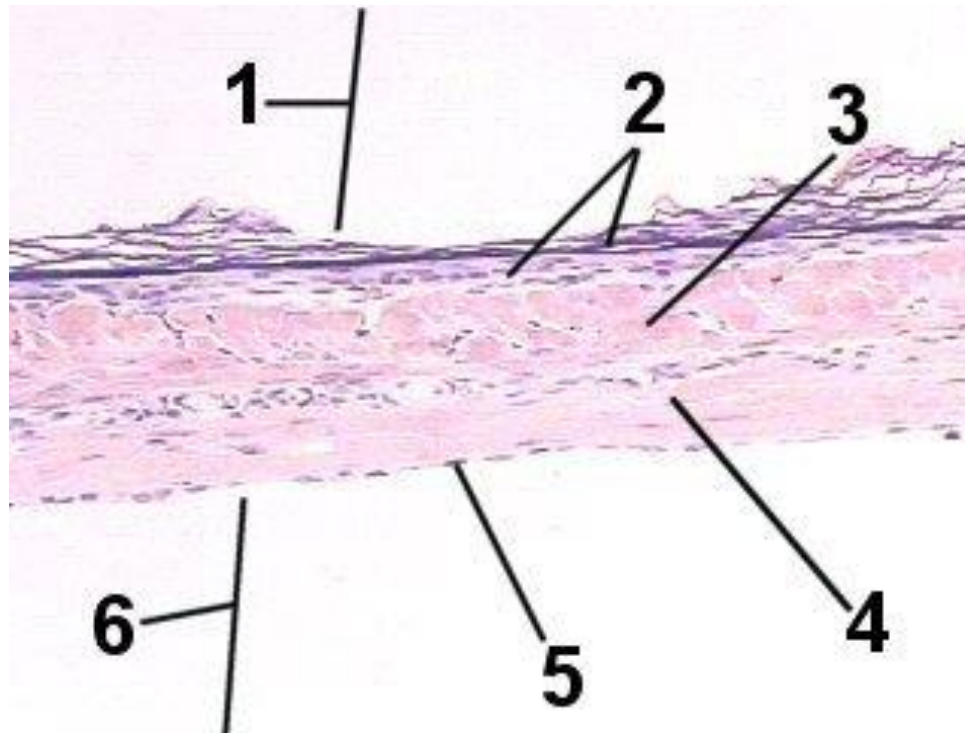
SCE = simple cuboidal epithelium

SSE = stratified squamous epithelium

Middle Ear

Tympanic Membrane

Photomicrograph x9, Inset x190

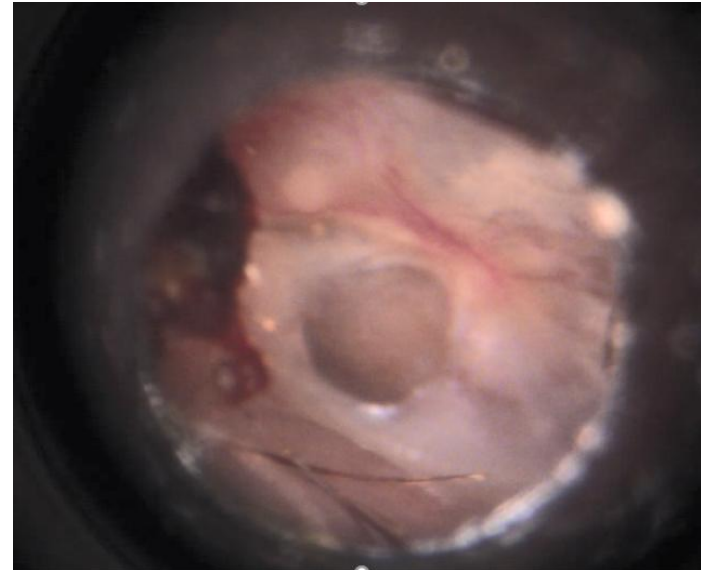
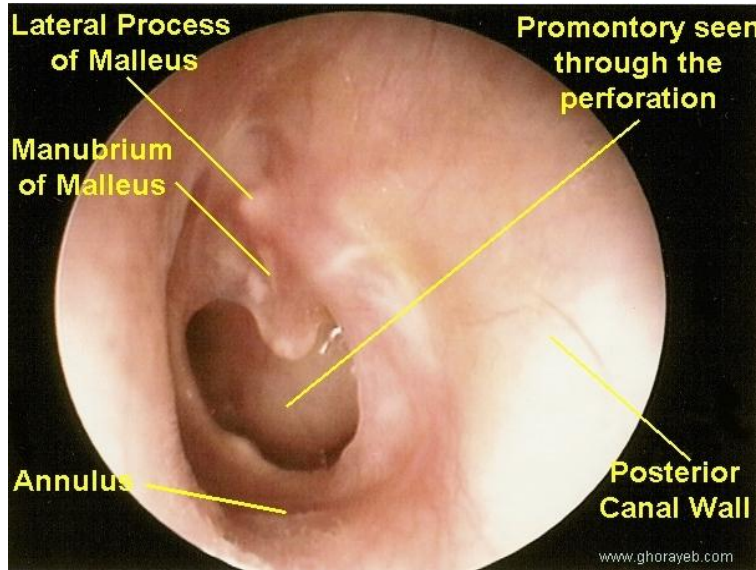


H&E high magnification

- | | |
|---|-----------------------------------|
| 1. External acoustic meatus | 4. Circular collagen fibers |
| 2. Thin skin : Keratinized stratified squamous epithelium | 5. Simple low cuboidal epithelium |
| 3. Radial collagen fibers | 6. Middle ear |

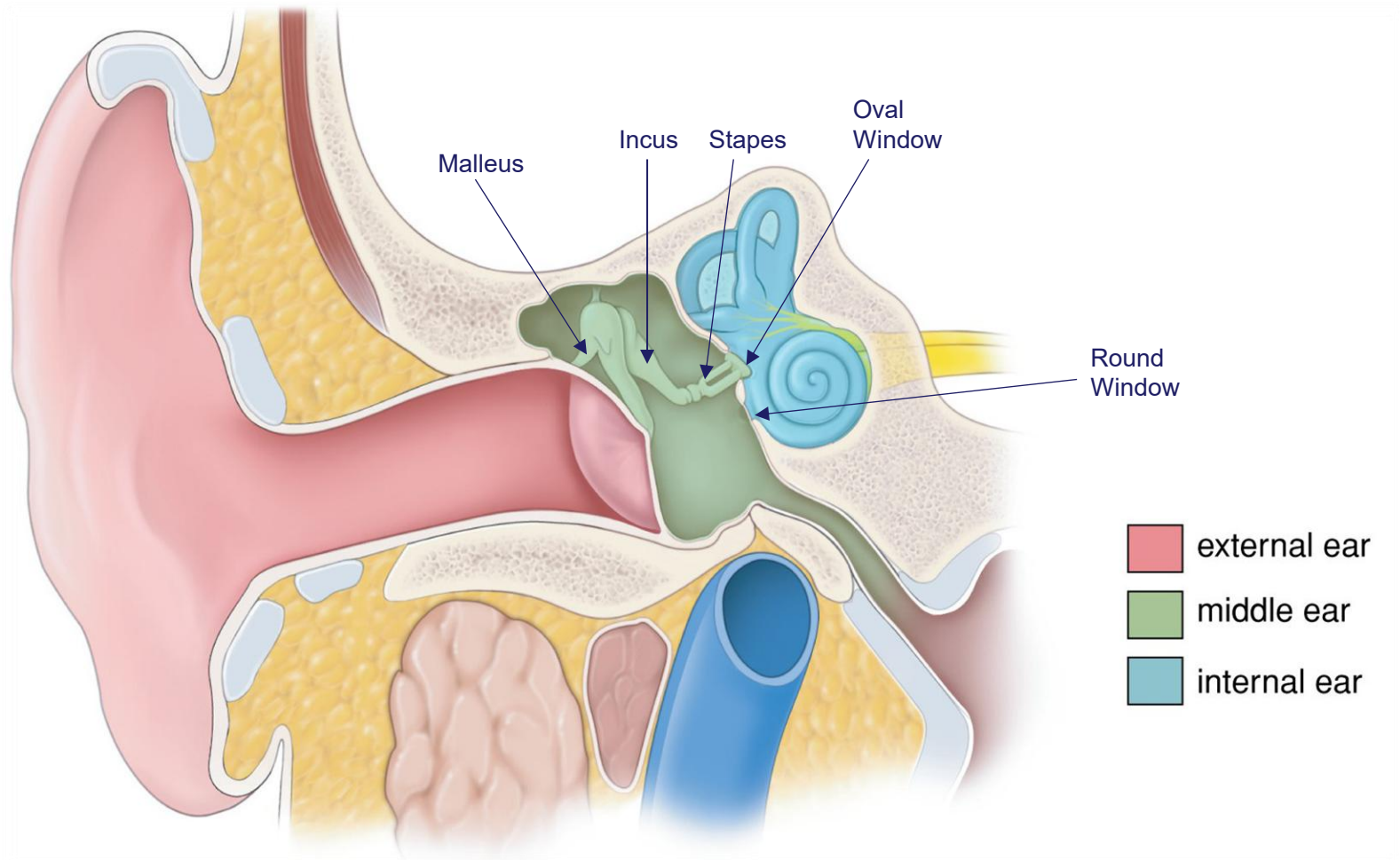
Middle Ear

Perforated Tympanic Membrane



- A rupture in the tympanic membrane.
- Creates a connection between the external acoustic meatus and the middle ear.
- Can be caused by infections, trauma, or loud noise.
- Symptoms include sudden ear pain, ear discharge, tinnitus, and vertigo.
- Most heal spontaneously, though some cause temporary or permanent hearing loss.

Middle Ear: Auditory Ossicles



- **The auditory ossicles connect the tympanic membrane to the oval window.**
 - Transmit and amplify mechanical vibrations and deliver them to the inner ear (via the oval window).

Internal Ear

Contains two labyrinthine compartments

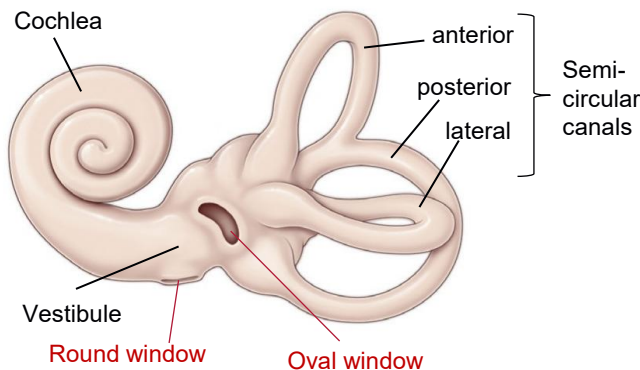


1. Bony Labyrinth

Three *connected* spaces:

- **3 Semicircular canals:** Anterior, posterior & lateral
- **Vestibule**
- **Cochlea**

Contains perilymph



2. Membranous Labyrinth

Communicating **sacs & ducts** suspended **within** the **bony labyrinth**. Two divisions:

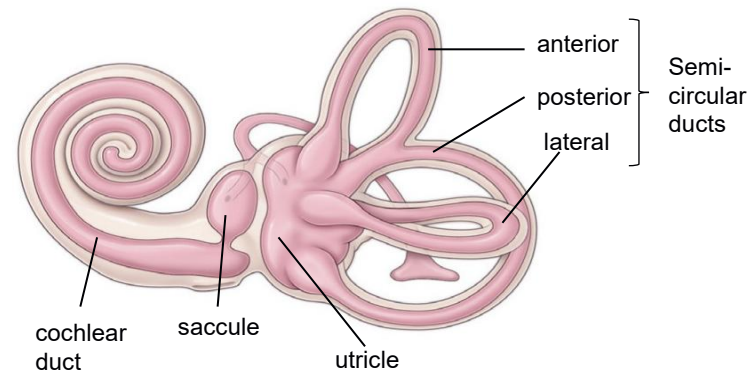
1. Vestibular labyrinth

- Semicircular ducts
- Utricle & Saccule

2. Cochlear labyrinth:

- Cochlear duct

Contains Endolymph



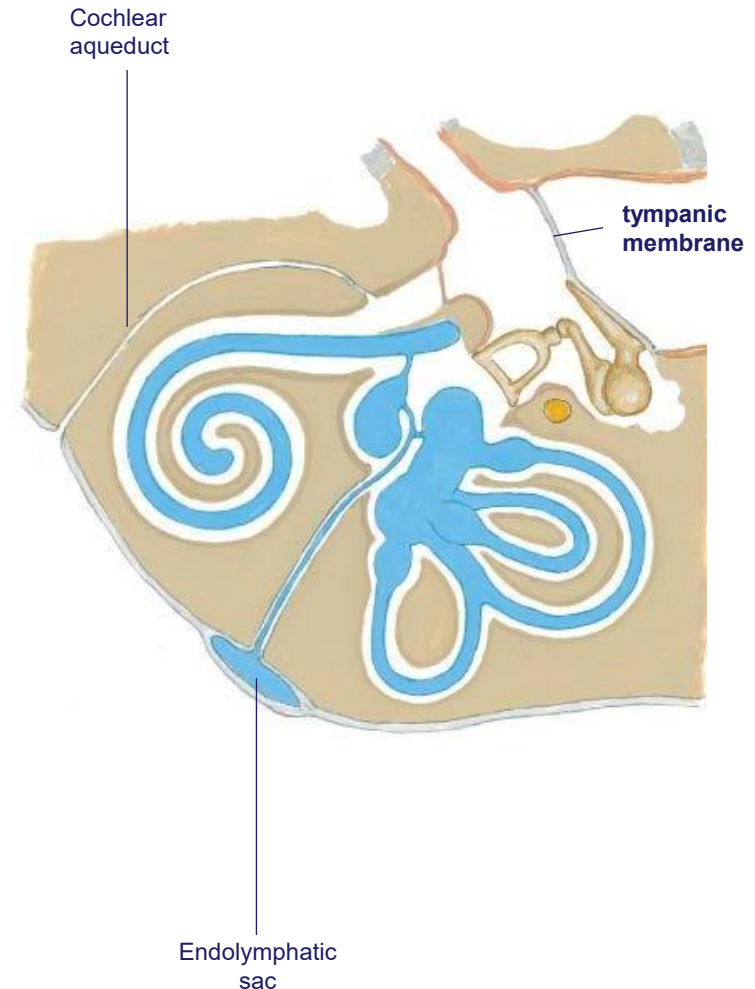
Perilymphatic & Endolymphatic Spaces

Perilymphatic space (white)

- Contained within the bony labyrinth.
- Perilymph
 - Similar in composition to extracellular & cerebrospinal fluid
 - ***Low K^+ /High Na^+ concentration***

Endolymphatic spaces (blue)

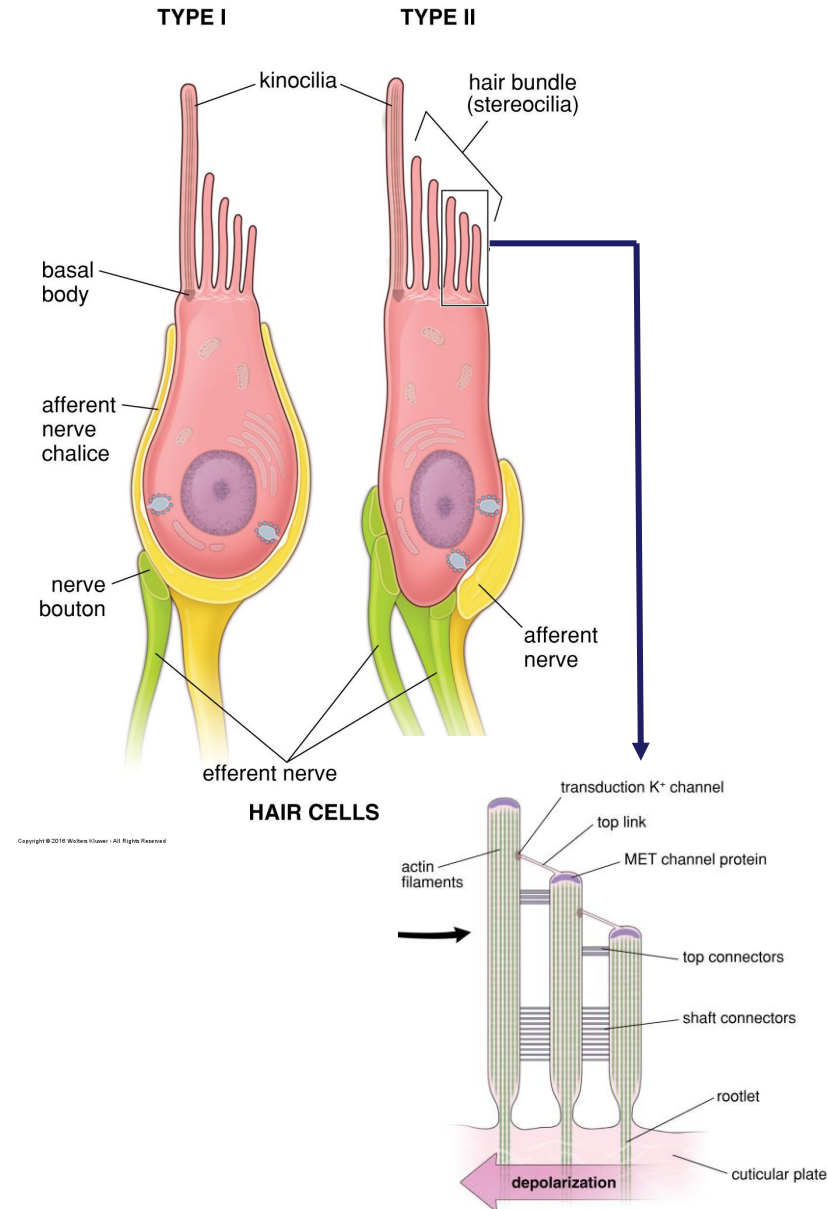
- Contained within the membranous labyrinth
- Endolymph
 - Similar in composition to intracellular fluid
 - ***High K^+ / Low Na^+ concentration***



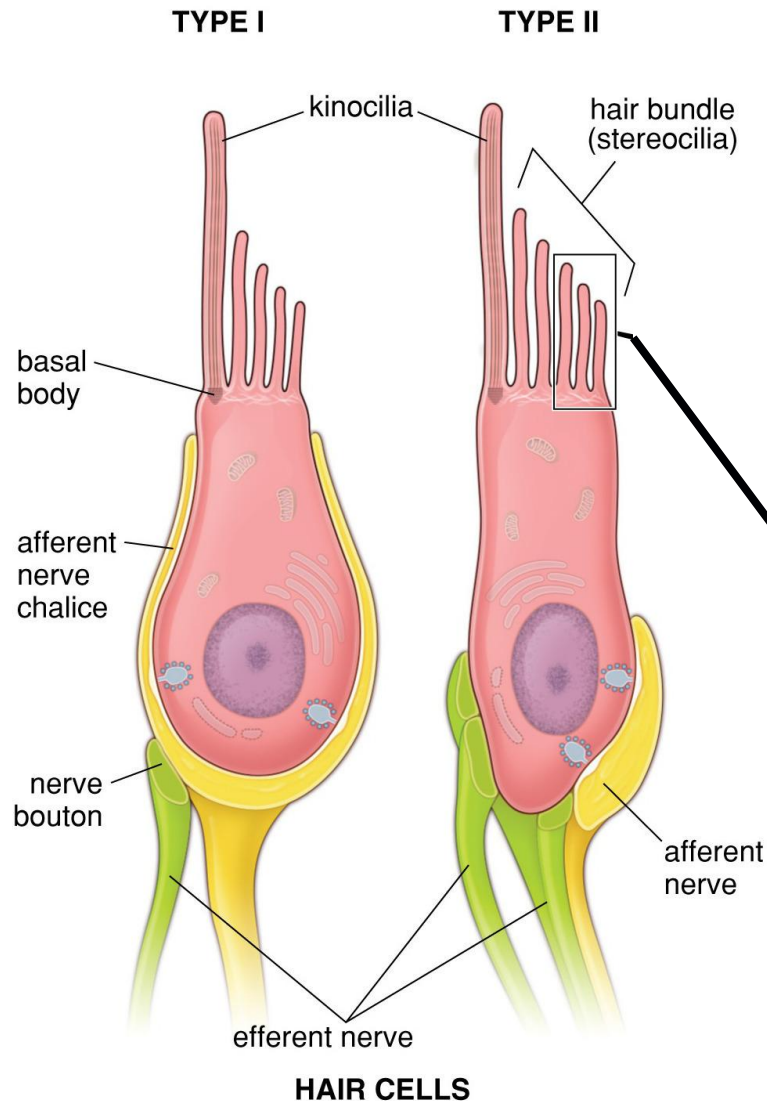
Sensory Cells of Membranous Labyrinth

Hair cells:

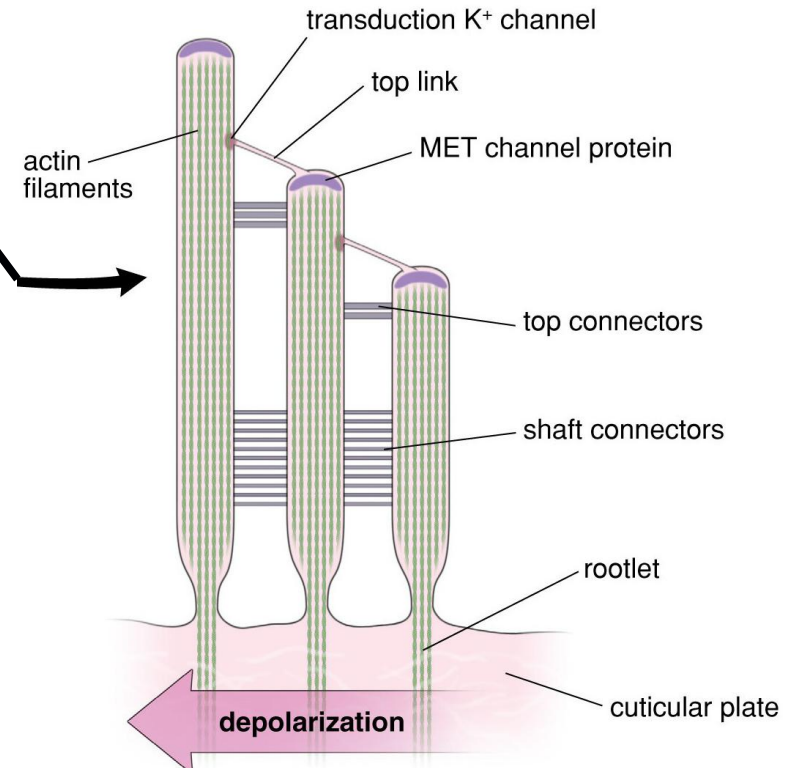
- **Mechanoreceptors** of the vestibular & cochlear labyrinths.
- Converts mechanical energy into electrical energy that is transmitted to the brain via the vestibulocochlear nerve.
- **Hair bundle:** rows of stereocilia, AKA Sensory hairs
 - Mechanoelectrical transducer (MET) channel proteins at the distal ends
- **Kinocilium**
 - **True cilium is found ONLY in the sensory regions of the vestibular system.**
 - Defines the polarity of the hair bundle
 - Positioned behind the longest row of stereocilia



Sensory Cells of Membranous Labyrinth



1. Deflection of stereocilia toward the tallest stereocilium increases top-link tension
2. This opens mechanically gated MET channels at the stereocilia tips
3. K^+ influx $\rightarrow$ hair cell depolarization
4. Depolarization opens voltage-gated Ca^{2+} channels at the basolateral membrane
5. Ca^{2+} influx triggers neurotransmitter release
6. Neurotransmitter stimulates adjacent auditory nerve fibers $\rightarrow$ action potentials



Sensory Regions of the Vestibular Labyrinth

1. Semicircular ducts

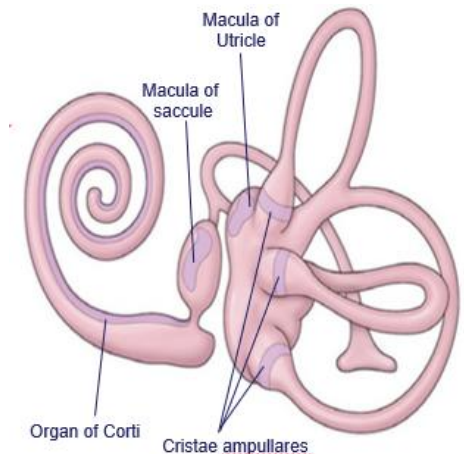
○ **Crista ampullaris**

- Sensors of the angular movement of the head
- Sensory Cells: Type I & II hair cells + supporting cells

2. Sacculle & utricle:

○ **Macula**

- Sensors of gravity & linear acceleration
- Sensory Cells: Type I & II hair cells + supporting cells



Clinical correlation: Vertigo

Sensory Regions of the Vestibular Labyrinth

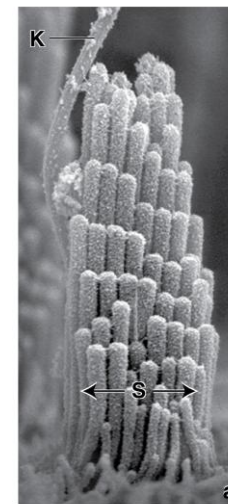
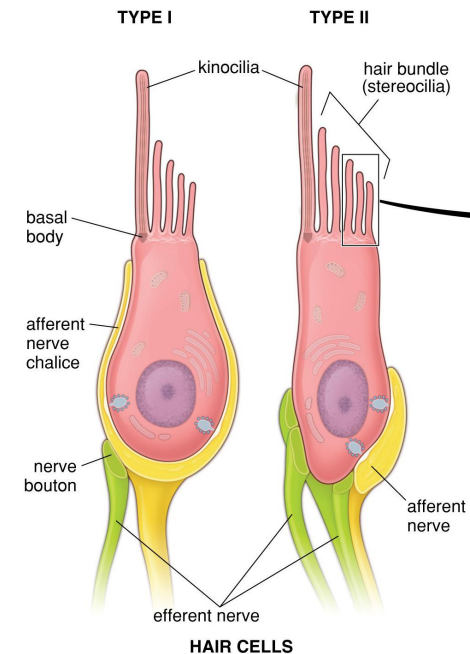
Sensory Cells:

○ Type I:

- Flask-shaped
- Surrounded by an afferent nerve chalice & a few efferent fibers

○ Type II

- Cylindrical shape
- Afferent & efferent bouton nerve endings



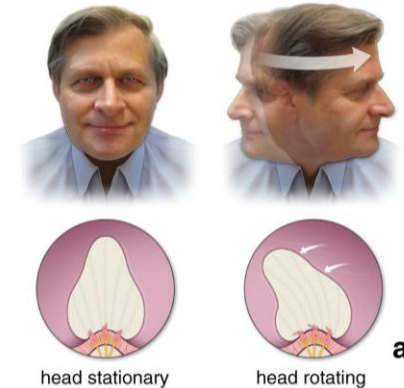
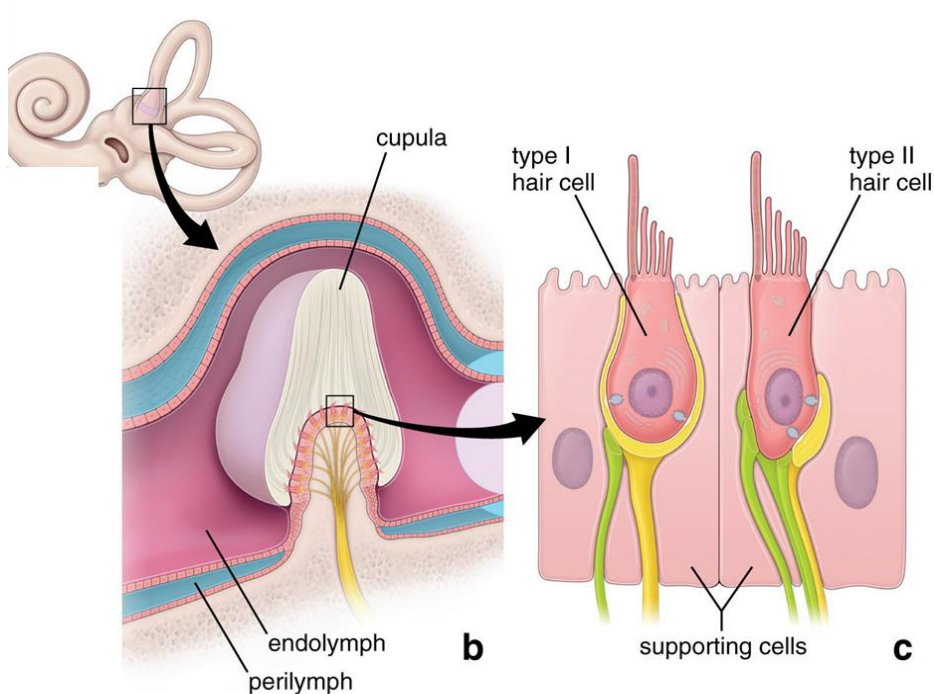
SEM
x47,500
K = kinocilium
S = stereocilia

TEM
x47,500
K = kinocilium
S = stereocilia

Note: All vestibular labyrinth cells have a kinocilium.

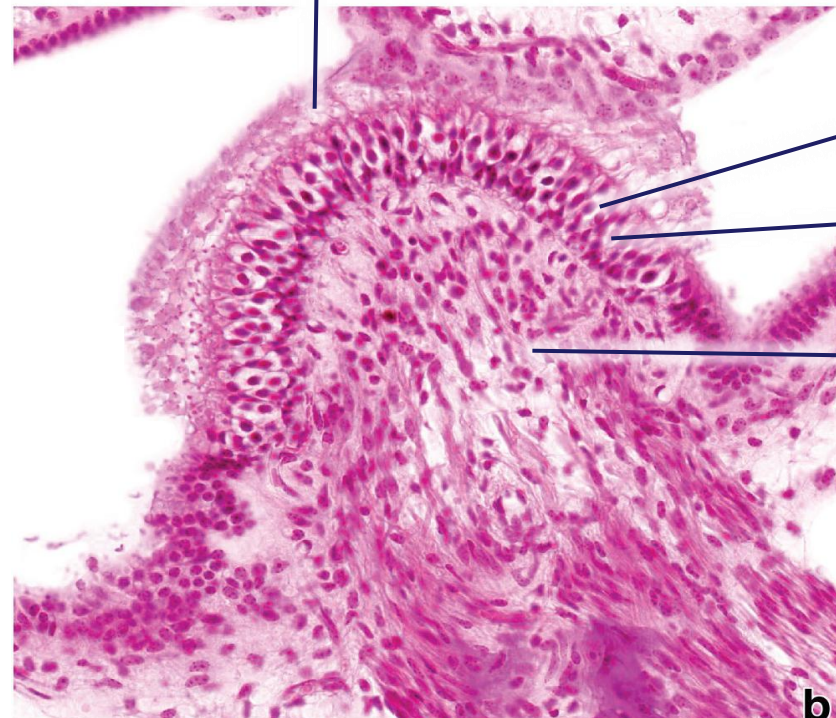
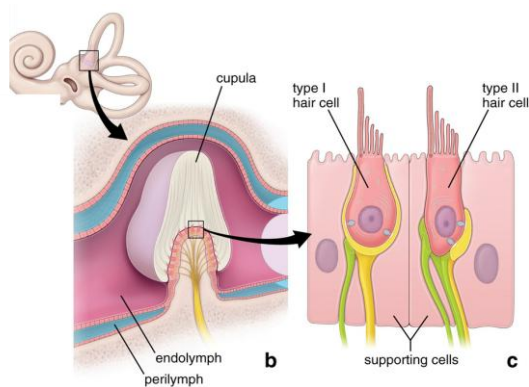
Crista Ampullaris of the Semicircular Duct

- **Sensory receptor for angular movement of the head. (turning the head left to right).**
- **Thickened epithelial ridge:**
 - **Sensory epithelium: type I and type II Hair cells** and support cells.
 - **Cupula:**
 - Gelatinous protein-polysaccharide mass
 - Projects into the lumen and is surrounded by endolymph.



1. Rotational movement of the head.
2. Semicircular canals and the membranous semicircular ducts move with head.
3. Movement of endolymph tends to be lagged.
4. Swaying of cupula in the opposite direction.
5. Deflection of stereocilia
6. Activation of MET channels
7. Generation of nerve impulse

Crista Ampullaris of the Semicircular Duct



Cupula

Hair cells

Supporting cells

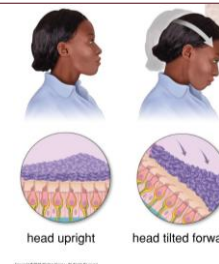
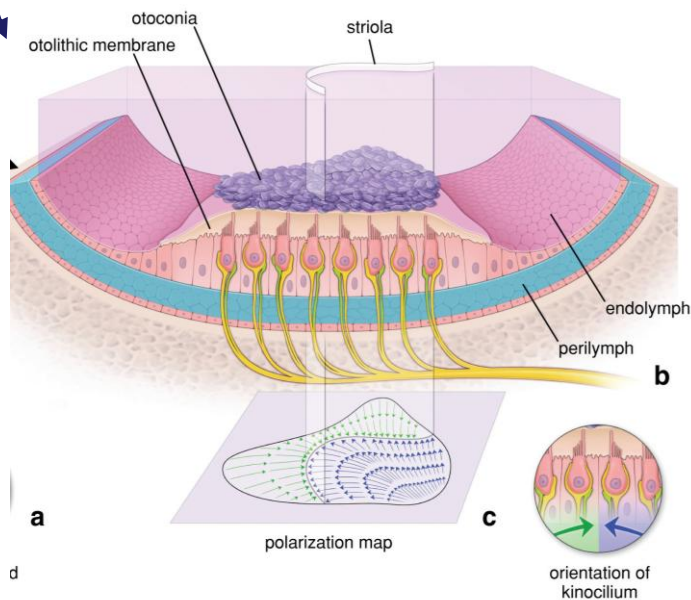
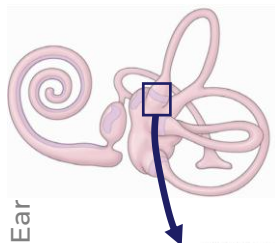
Nerve fibers

Sensory epithelium

b

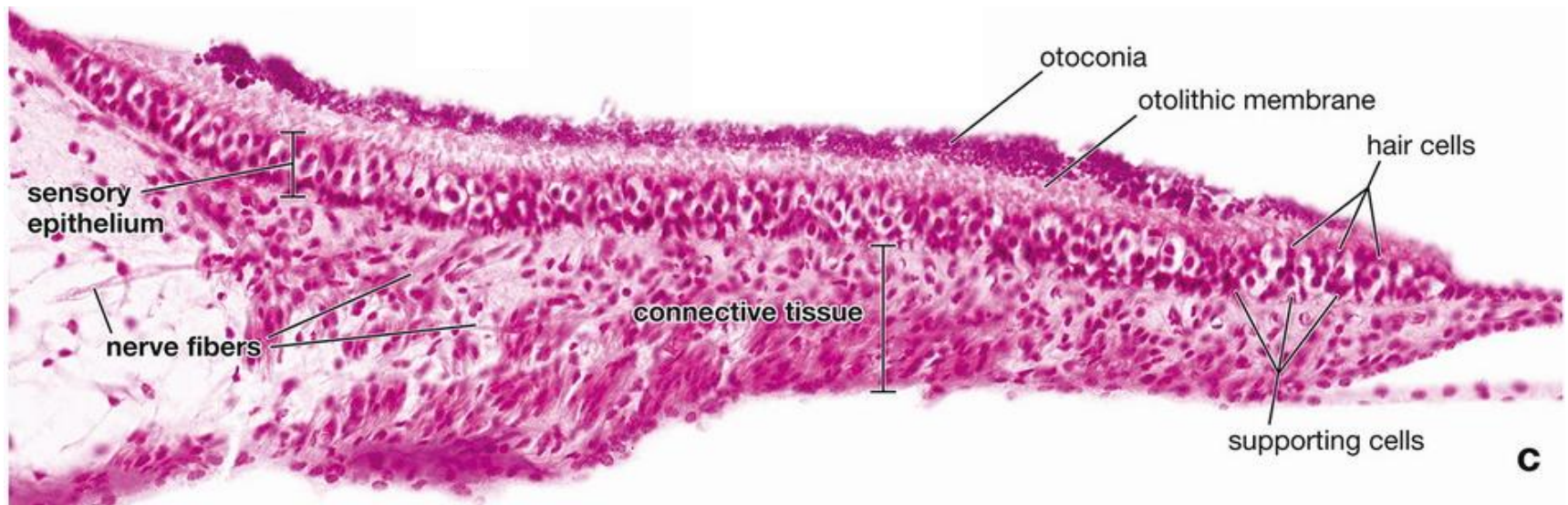
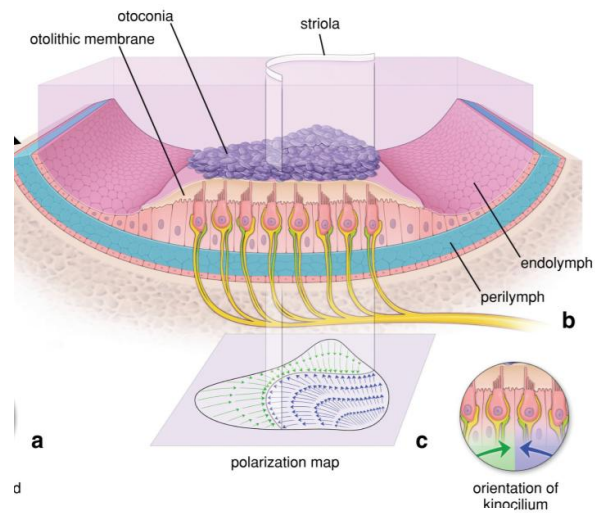
Macula of Utricle and Sacculle

- Sensory region of the Utricle and Sacculle, respectively, which are part of the vestibular labyrinth.
- **Sensory receptors for gravity & linear acceleration (moving forward/backward or up/down, or when gravity acts on your head (tilting))**
- Thickened epithelial ridge:
 - **Sensory epithelium: type I and type II Hair cells** and supporting cells
 - Polarized to the striola, allowing detection of movement in multiple directions.
 - **Otolithic membrane:**
 - Gelatinous polysaccharide mass that the hair cell processes are embedded in.
 - Outer surface covered in otoliths/otoconia: Crystalline bodies of calcium carbonate.

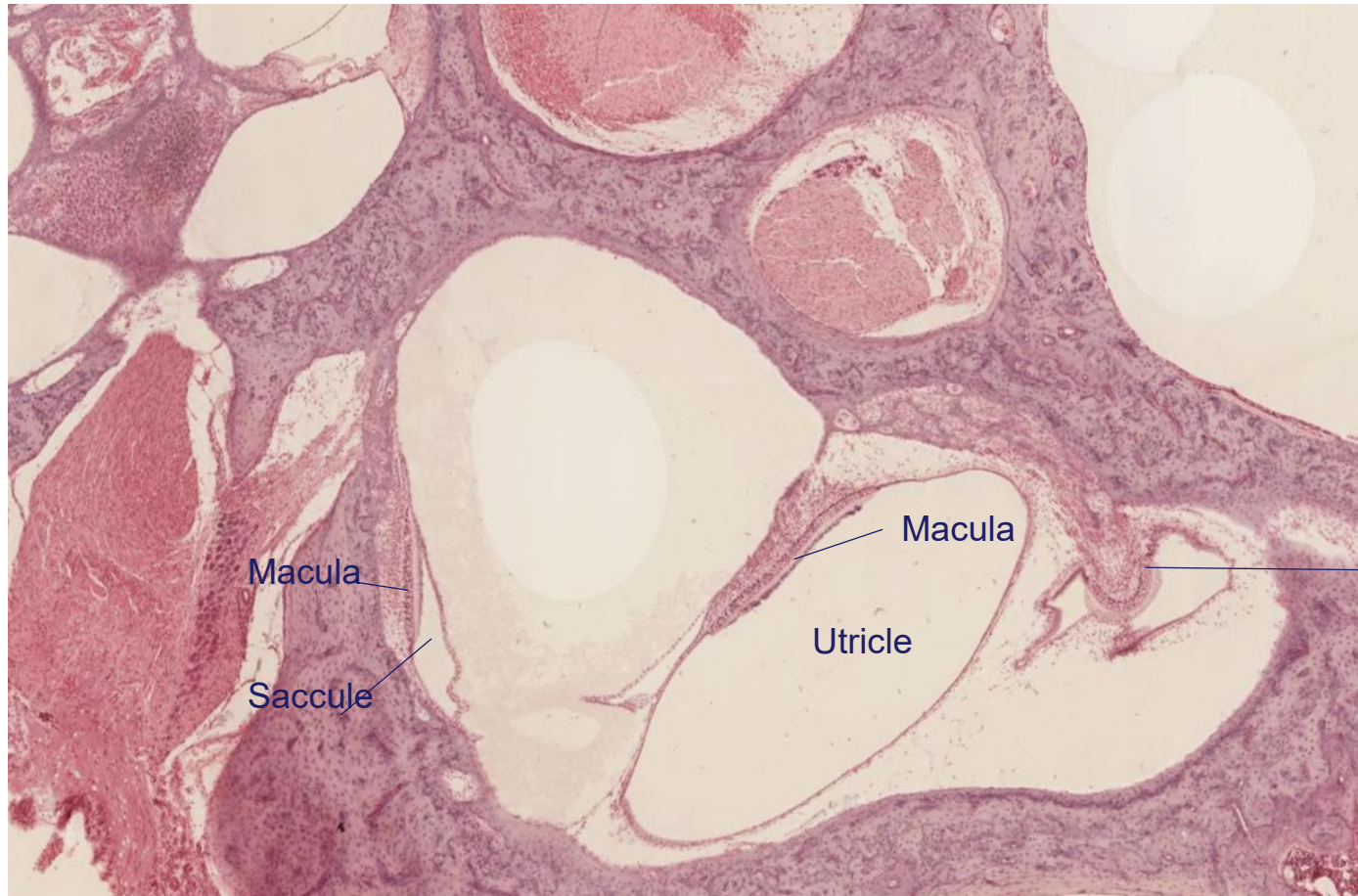


1. Linear acceleration or head tilt
2. Movement of endolymph
3. Otoconia lag and shift or pull on the otolithic membrane
4. Deflection of stereocilia
5. Activation of MET channels
6. Generation of nerve impulse

Macula of Utricle and Sacculle



Vestibular Labyrinth



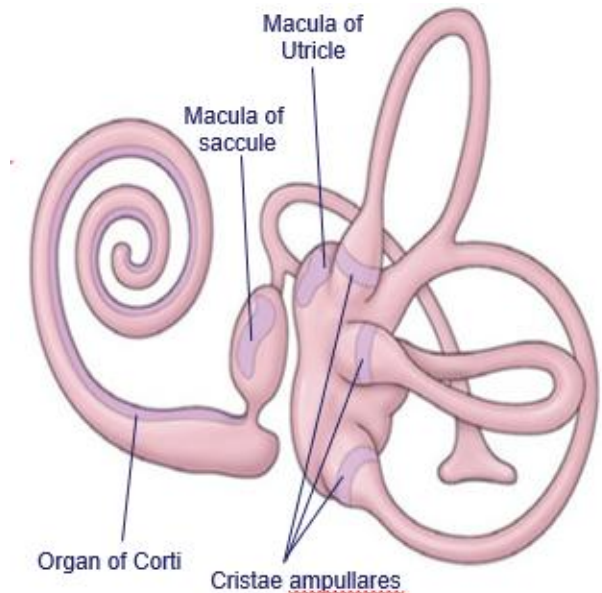
Ear Cochlear duct, utricle (macula) and ampulla (crista) H&E Copyright Regents of the University of Michigan

Sensory Regions of the Cochlear Labyrinth

1. Cochlear Duct

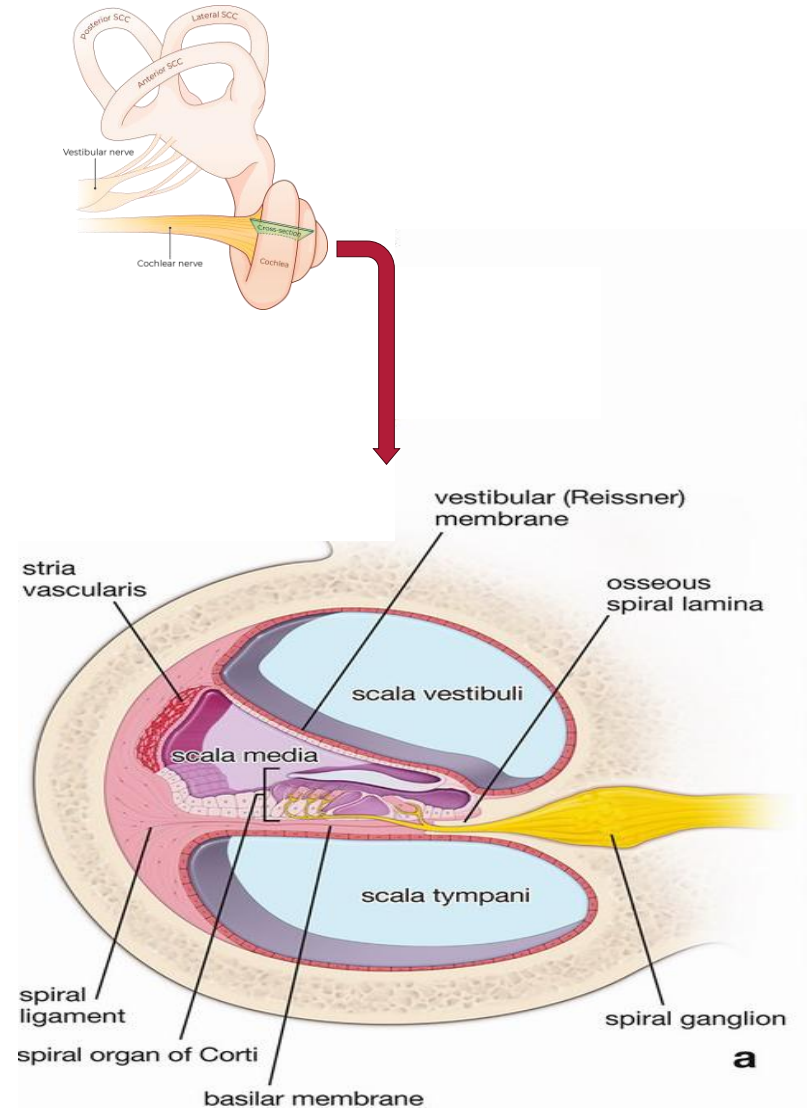
- **Spiral organ of Corti**
- Sensor of sound vibrations
- Sensory cells: Inner & outer hair cells + supporting cells
- Spiral ganglion: cell bodies of the neurons of the cochlear nerve.

Clinical correlation: Sensorineural hearing loss

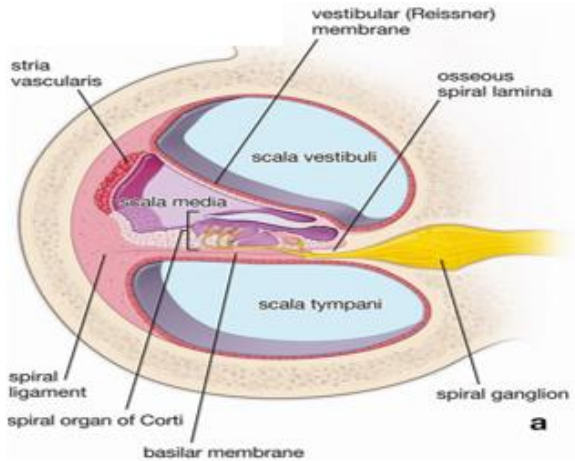


Cochlear Labyrinth

- Contains the cochlear duct that coils $2\frac{3}{4}$ turns around the **modiolus**.
 - **Modiolus**: the central bony core of the cochlear that contains nerves and blood vessels.
- It divides the cochlear canal into **3 compartments**:
 1. **Scala vestibuli** → Perilymph
 2. **Scala media (cochlear duct)**
 - Endolymph space
 - Contains the spiral organ of Corti (sensory region)
 3. **Scala tympani** → Perilymph
- Scala vestibuli & scala tympani connect at the apex of the cochlea via a small channel called the **helicotrema**.

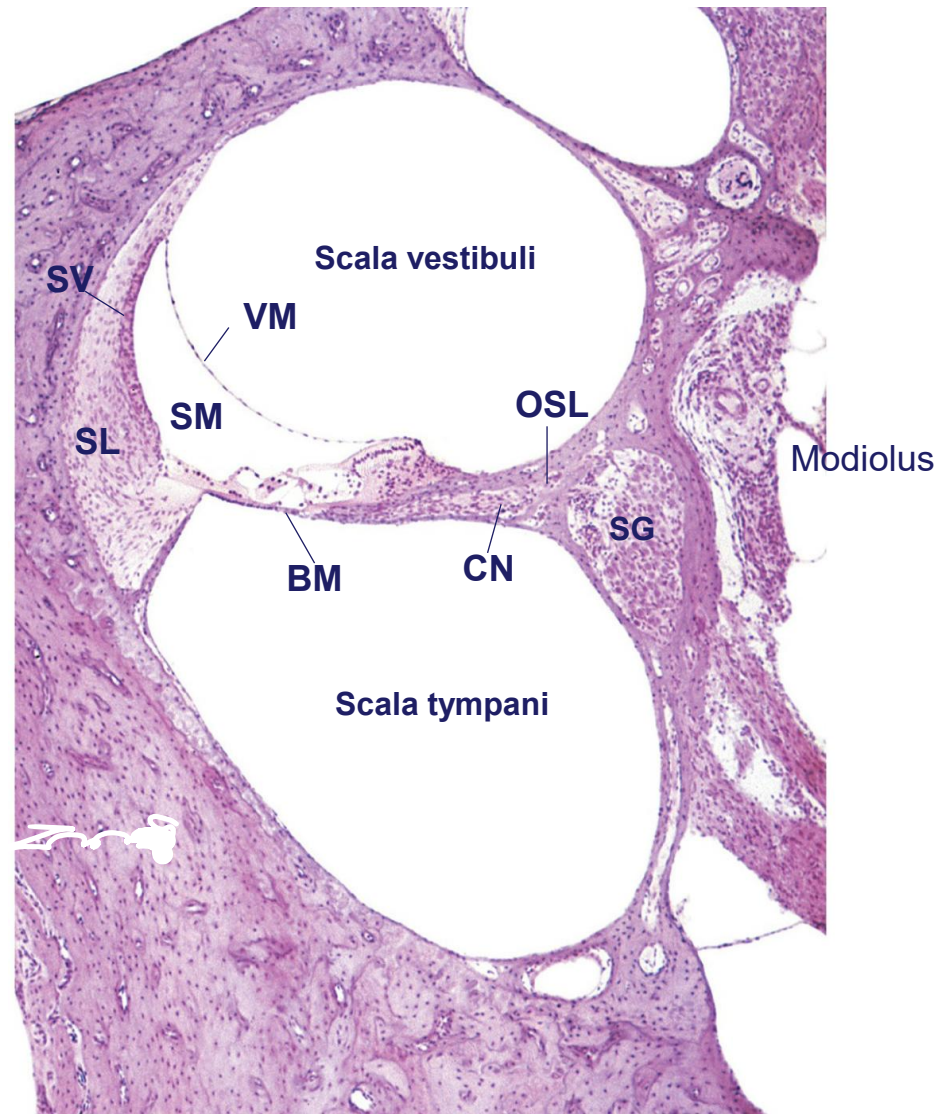


Cochlear Labyrinth



SM = scala media

- **SV = stria vascularis** produces and maintains endolymph
- **VM = vestibular (Reissner's) membrane** separates the scala vestibular from the scala media
- **OSL = osseous spiral lamina** is the bony continuation of the modiolus
- **BM = basilar membrane** is membranous continuation of the osseous spiral lamina.
 - **Spiral Organ of Corti sits on the basilar membrane**



H&E x65

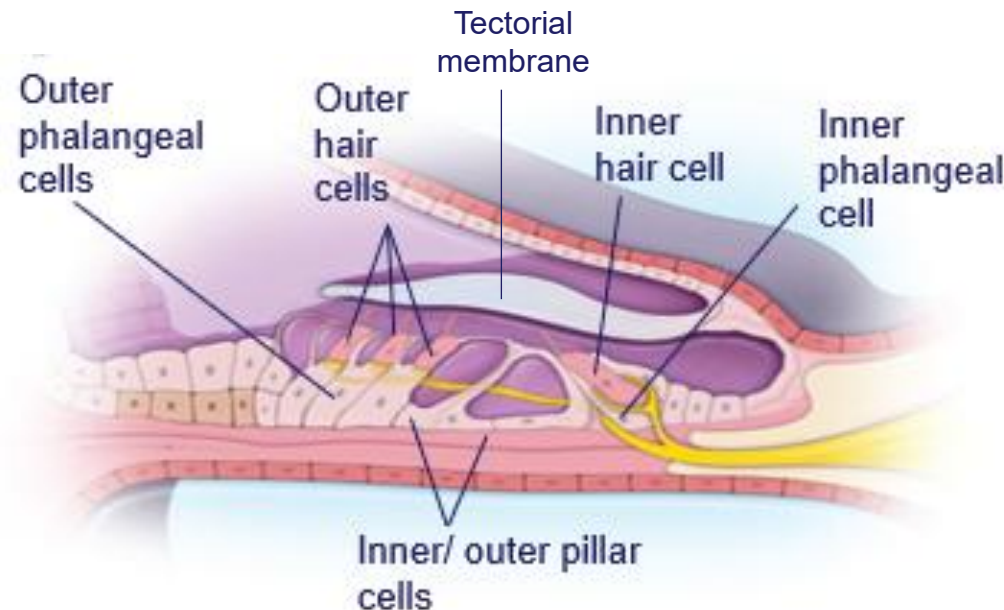
SG = spiral ganglion of cochlear nerve

CN = dendrites of cochlear nerve

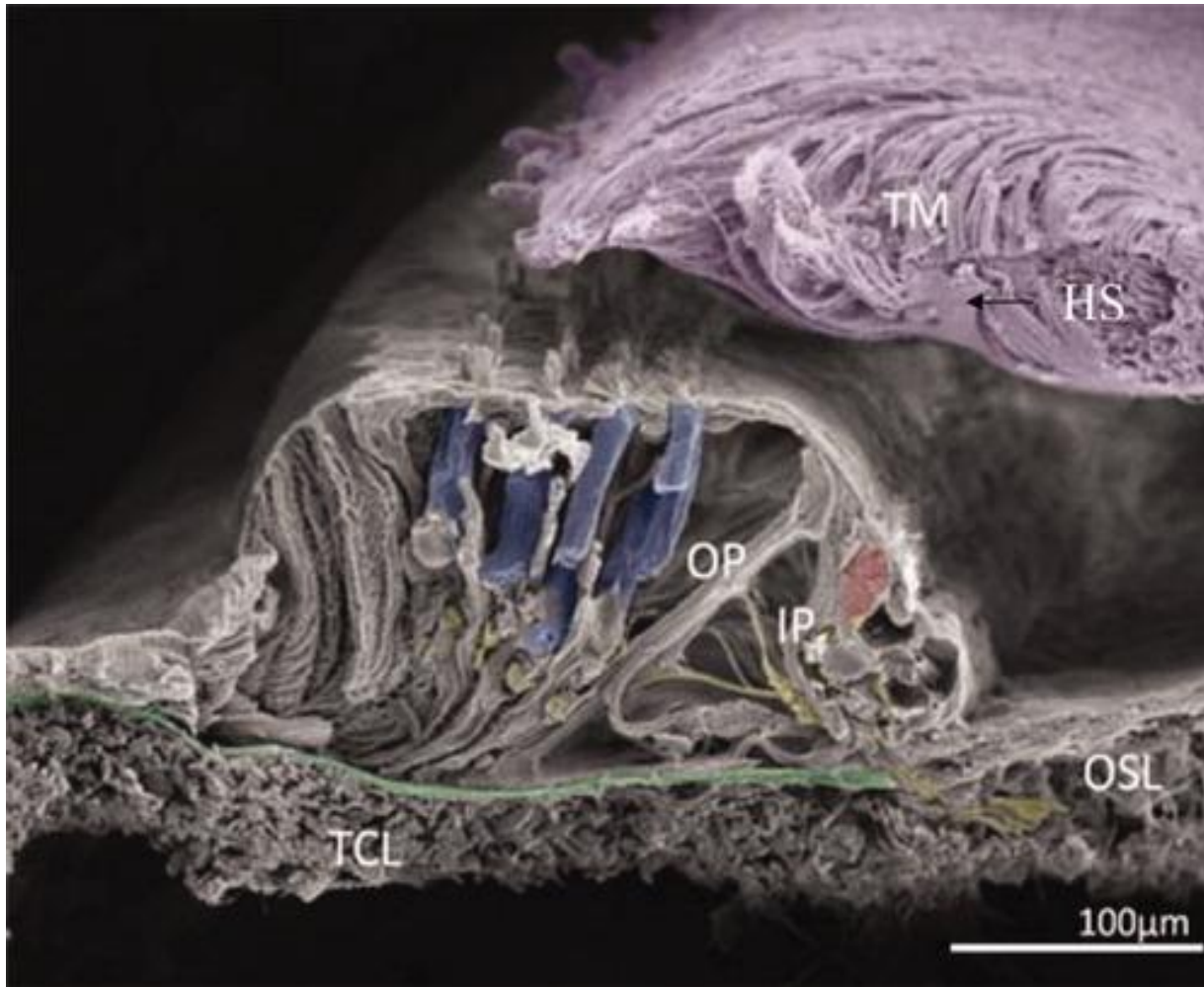
SL = spiral ligament

Spiral Organ of Corti

- Complex **epithelial layer on basilar membrane** (floor of the scala media).
- Overlain by the **tectorial membrane**
 - Gel-like structure: collagen II, V & IX; glycoproteins
 - Attached medially to the modiolus
 - Lateral edge projects over & attaches to the stereocilia of hair cells.
- Composed of **Hair Cells** (Inner and Outer), **Phalangeal Cells** (Inner and Outer), **Pillar cells**



Spiral Organ of Corti

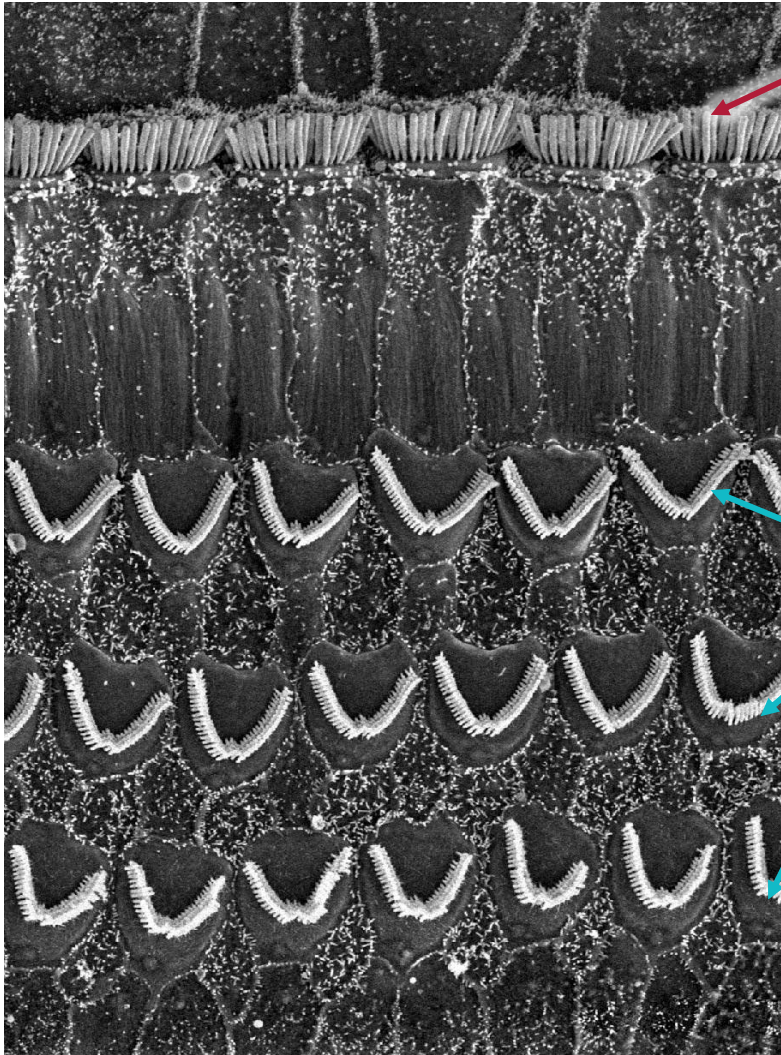


Human Cochlea: Anatomical Characteristics and their Relevance for Cochlear Implantation, Volume: 295, Issue: 11, Pages: 1791-1811, First published: 08 October 2012, DOI: (10.1002/ar.22599)

SEM human cochlea

- Outer hair cells (blue)
- Inner hair cells (red)
- Basilar membrane (green)
- Tectorial membrane (TM)
- Neurons (yellow)

Hair Cells of the Spiral Organ of Corti



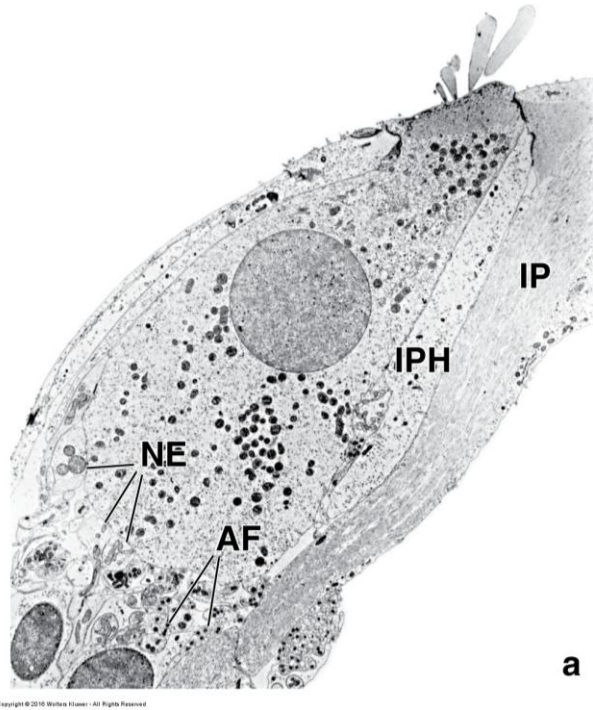
Inner hair cells

- Single row of cells along the cochlear duct
- Straight line of stereocilia
- **Primary sensory cells for hearing**
- Receive ~90% of auditory nerve fibers
- Do not regenerate

Outer hair cells

- Multiple rows (usually 43; up to 5 at apex)
- Stereocilia arranged in “W” pattern
- Receive ~10% of auditory nerve fibers
- **Function mainly in sound amplification**

Inner & Outer Hair Cells



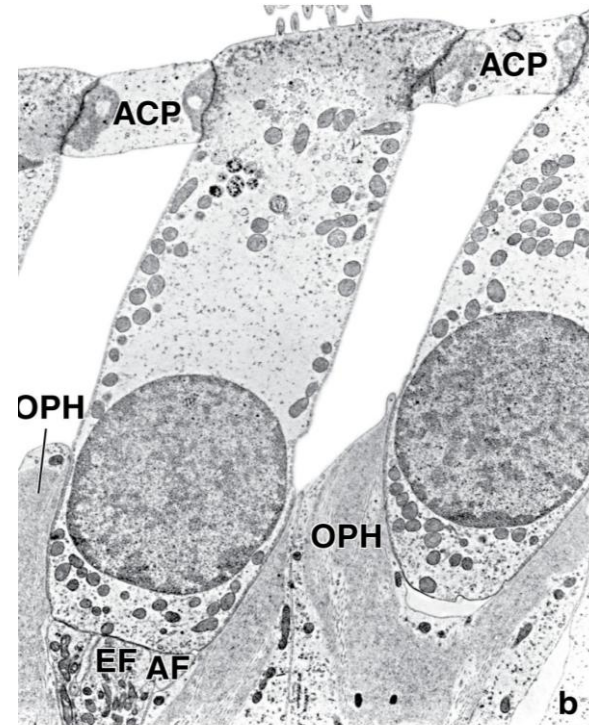
a. Inner hair cell

NE = nerve endings

AF = afferent fibers

IPH = inner phalangeal cell

IP = inner pillar cell



b. Outer hair cells

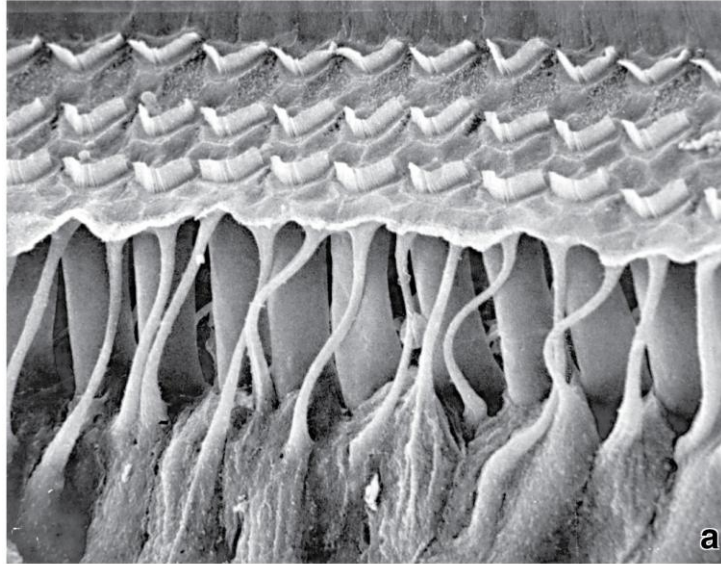
ACP = apical cuticular plate

OPH = outer phalangeal cell

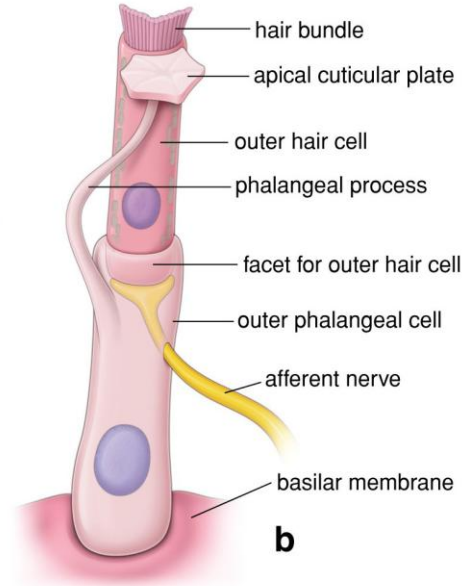
AF = afferent nerve fibers

EF = efferent nerve fibers

Outer Phalangeal Cell



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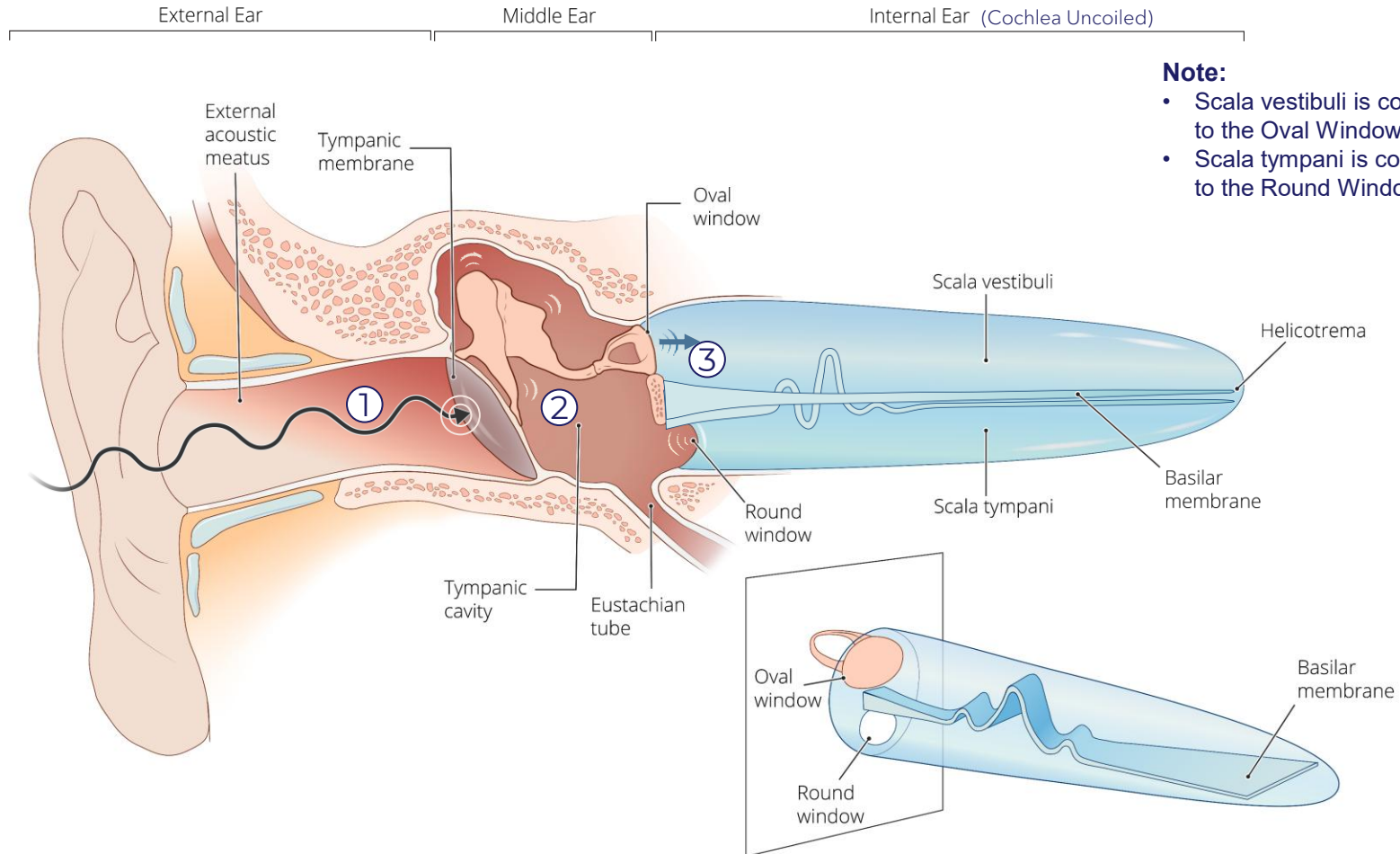


SEM X 2400

- Supports outer hair cells
- Apical processes form plates around the hair cells
- Plates are joined by tight junctions and form the **reticular lamina**
- Reticular lamina separates **endolymph** from intercellular spaces
- Intercellular spaces contain **cortilymph (Extracellular fluid produced by supporting cells)**

Sound Perception

Illustrations provided by the Center for Biomedical Visualization (SGU)



Note:

- Scala vestibuli is connected to the Oval Window
- Scala tympani is connected to the Round Window

1. **Sound waves** enter the external ear and reach the middle ear, where they are converted into **mechanical vibrations**.
2. These vibrations are passed to the oval window and converted into fluid vibrations/waves
3. The fluid waves move through the perilymph of the scala vestibuli, then across the vestibular membrane to the scala media (with endolymph) and to the scala tympani.
 - **These fluid waves/vibrations cause displacement of the basilar membrane creating a traveling wave.**

Sound Perception

Sound waves cause vibration of the **basilar membrane**, moving the Organ of Corti

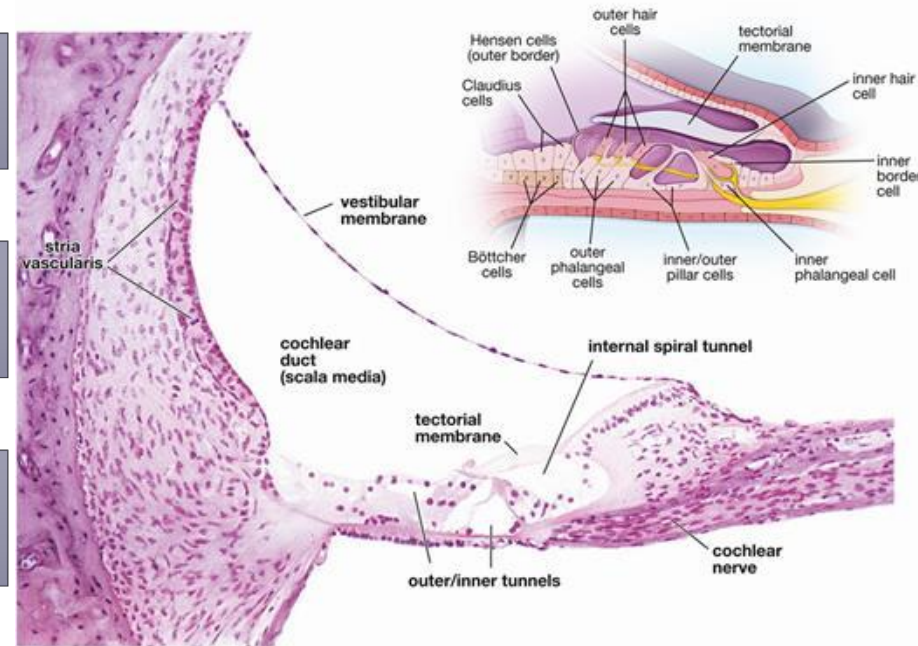
Relative motion between the **basilar membrane** and **tectorial membrane** causes a shear force.

Shearing deflects the stereocilia on the cochlear hair cells.

Deflection opens **mechanically gated MET channels**, causing hair cell depolarization.

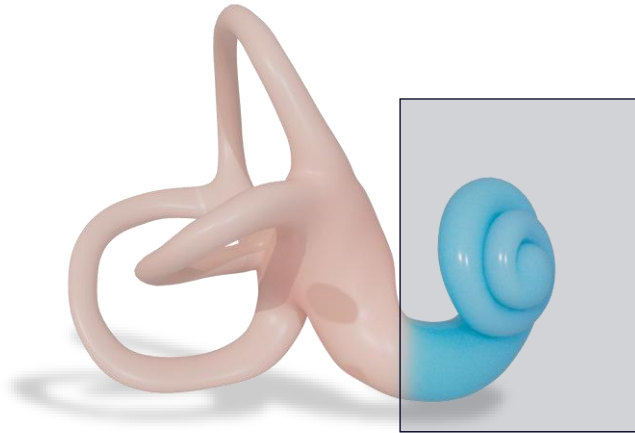
Depolarized hair cells release neurotransmitters onto afferent fibers of the cochlear nerve.

Afferent neurons generate action potentials that travel to the brain via the cochlear nerve.



Traveling Wave Frequencies

Cochlea (uncoiled)

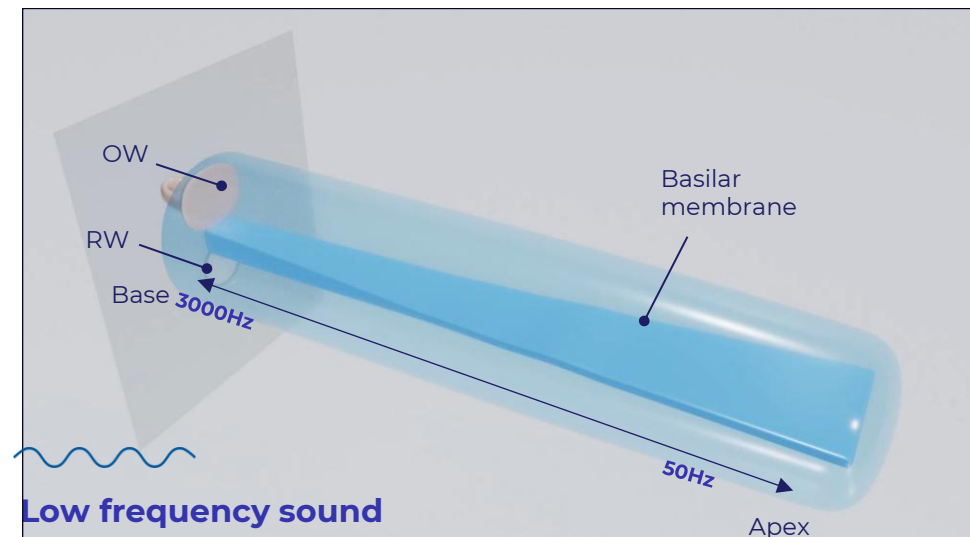
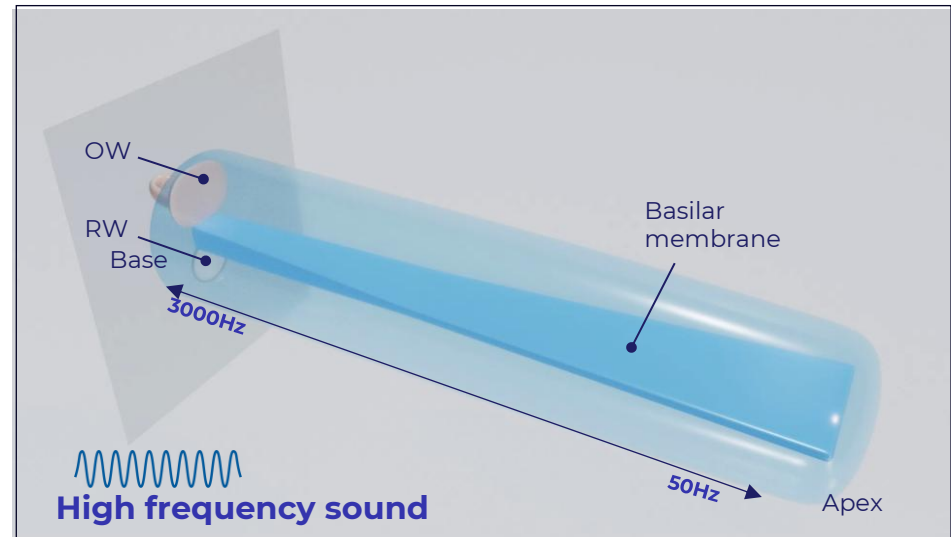


Cochlea

Sound originating from oval window (**OW**) and traveling to round window (**RW**).

Different frequencies cause maximum displacement/vibrations of certain segments along the length of the basilar membrane

- Basilar membrane is relatively narrow & thick at the base - **High frequency sounds**
- Basilar membrane is relatively wide and thin at the apex - **Low frequency sounds**



Clinical Correlation – Hearing Loss

• Conductive hearing loss

- Sound waves are mechanically impeded from reaching the auditory sensory receptors of the internal ear
- Causes
 - Ear infection (otitis media)
 - Excess ear wax
 - Foreign bodies
 - Otosclerosis

• Sensorineural hearing loss

- Injury to the sensory receptors, nerves or auditory cortex
- Causes
 - Infections of membranous labyrinth
 - Fracture of temporal bone
 - Acoustic trauma
 - Prolonged exposure to excessive noise
 - Aging
 - Presbycusis
 - Loss of sensory cells
 - Begins at basal end of cochlea
 - High frequency hearing loss

Frequency of Sounds (measured in hertz (Hz))

Vowel Sounds Like a Short "O"

250 - 1,000 hz

Consonant Sounds Like "S"

1,500 - 6,000 hz

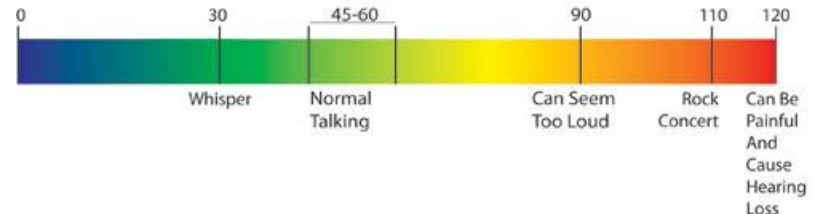
Important Sounds

250 - 6,000 hz

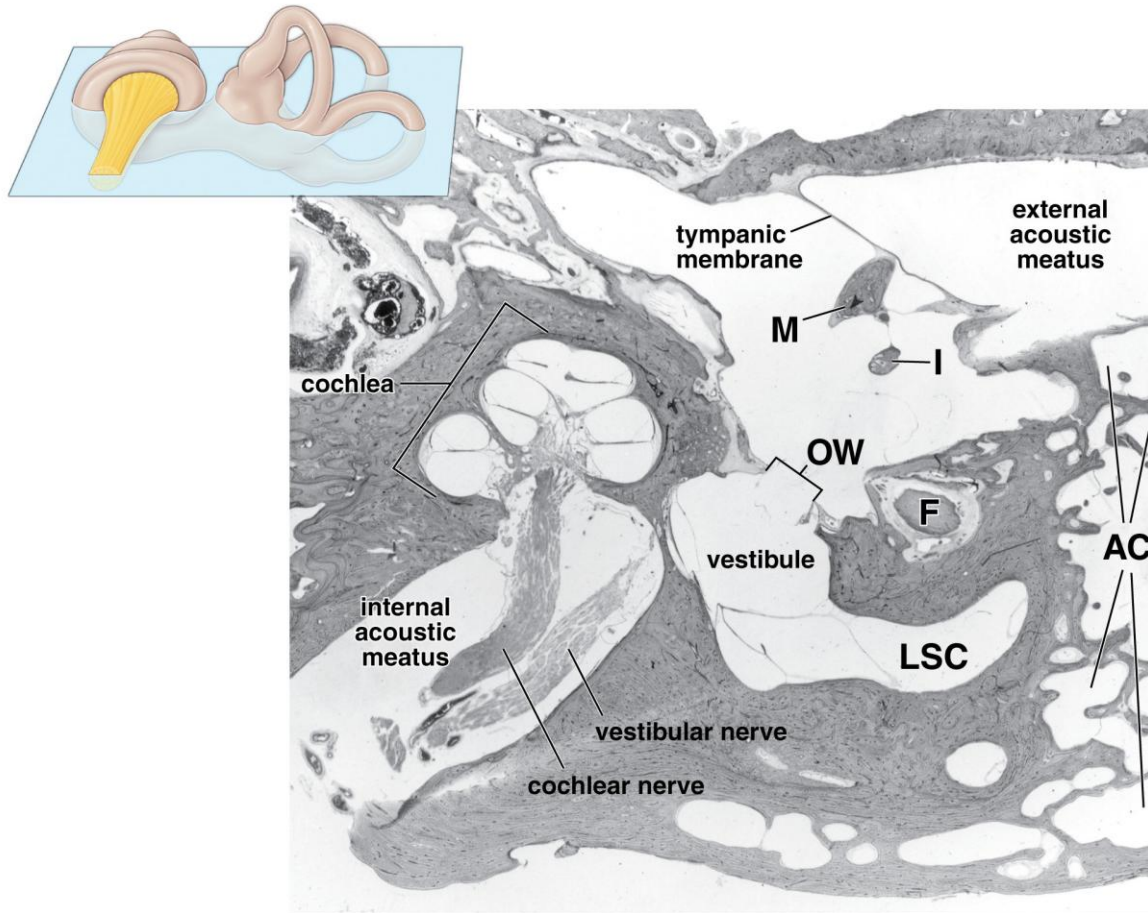
Normal Hearing

20 - 20,000 hz

Intensity of Sounds (Measured in Decibels (dB))



Horizontal Section of Temporal Bone



Photomicrograph x65

M= malleus

I= incus

OW= oval window

LSC= lateral semicircular canal

F= facial nerve

AC= mastoid air cells

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Basic Principles of Medicine II

Module: Neuroscience & Behavioral Sciences| Lecture NB 21

Thank you!!!

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